Neural Network Parameter Estimation for Discrete Bayesian Networks with Applications in Medical Diagnosis

Kaylin Aspoas
3163666

Supervised by Prof. Ritesh Ajoodha & Aashna Gaogle



UNIVERSITY OF THE
WITWATERSRAND,
JOHANNESBURG



The School of Computer Science and Applied Mathematics
The University of the Witwatersrand, Johannesburg
South Africa

A research proposal submitted to the Faculty of Science in partial fulfilment of the degree Honours of Science in Computer Science

May 20, 2026

Abstract

Bayesian networks offer an interpretable framework well-suited to high-stakes fields such as medical diagnosis. However, their standard parameter estimation methods, such as Maximum Likelihood Estimation and Bayesian Parameter Estimation, fail to capture the complex, non-linear relationships that characterize high-dimensional, sparse datasets typical of these domains. These methods become computationally infeasible as parent node in-degree increases.

This research investigates whether neural networks can replace traditional conditional probability tables for local parameter estimation within discrete Bayesian networks, preserving their interpretable graphical structure while improving representational capacity. Four neural conditional probability distribution architectures (ranging from a linear softmax baseline to a deep multilayer perceptron) are compared against traditional Maximum Likelihood Estimation. Experiments are conducted on synthetic datasets sampled from ground-truth networks of varying size, density, and parent in-degree. This enables direct evaluation of parameter estimation methods across increasing levels of network complexity, by comparing them to the ground-truth distributions using Kullback-Leibler divergence. The framework is then applied to a real clinical dataset, where estimator performance is assessed by held-out predictive log-likelihood.

This work provides an analysis of where and how standard parameter estimation techniques fail over high-dimensional, complex structures, such as those observed in clinical settings. Utilizing these insights, a neural parameter estimation framework is constructed that is expressive and compatible with the interpretable structure that makes Bayesian networks well-suited to clinical decision support.

Keywords: Bayesian networks; Parameter Estimation; Neural Networks;

Declaration

I, **Kaylin Aspoas**, declare that this research report is my own, unaided work. It is being submitted in partial/full fulfillment of the requirements for the degree Honours of Science in Computer Science and Applied Mathematics at The University of the Witwatersrand, Johannesburg.

This research has not been submitted before for any degree or examination at any other university. Where the work of others has been used, proper acknowledgment has been given.

Kaylin Aspoas
Student Number: 3163666

Signature: 

Date: 25/09/2026

Wits University Faculty of Science
Postgraduate Student AI Declaration

I understand that the use of generative AI tools (such as ChatGPT or similar) without explicitly declaring such use constitutes a form of plagiarism and is classified by Wits University as academic misconduct.

I declare that in the course of conducting the research towards my degree or in the preparation of this **thesis/dissertation/research report** (select one by marking with an **X**):

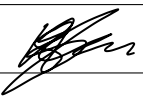
- ❑ I did not make use of generative AI tools
- ☑ I did make use of generative AI tools for the following (tick all that apply):
  - ❑ Idea Generation (research problem/design, hypothesis)
  - ☑ Sourcing Related Work (summarising, identifying sources)
  - ❑ Methods and Experiment Design (experiment setup, model tuning)
  - ❑ Data Analysis (presentation, coding, interpretation)
  - ❑ Theoretical Development (theorem proving, conceptual analysis)
  - ❑ Code Development (generating algorithms, writing scripts)
  - ☑ Presentation (rendering graphics, formatting)
  - ☑ Editing (grammar, readability)
  - ❑ Writing (text generation, document structuring)
  - ☑ Citation Formatting (structuring, organising)

If other uses were involved, please specify below:

Generative AI tool used (list all)	**Used for?**

If generative AI tools were used as an integral part of the experimental design or in the direct execution of my research, I confirm that details of this use are clearly outlined in the relevant experimental/methodology chapters of my thesis/dissertation/research report.

Student number: 3163666

Candidate signature: 

Date: 25/09/2026

Contents

1 Introduction **2**

2 Related Work **3**

2.1 Conceptual Framework 3
2.2 Interpretable Models for Medical Diagnosis 4
2.3 Parameter Estimation for Discrete Bayesian Networks in Medical Contexts . . . 5
2.4 Hybrid Models for Interpretability and Predictive Accuracy 6
2.5 Research Gap 7

3 Methodology **8**

3.1 Problem Statement 8
3.2 Purpose Statement 9
3.3 Research Questions and Experimental Setup 9
3.4 Detailed Methodology 10
3.4.1 Data Generation and Pre-processing 10
3.4.2 Methods 12
3.4.3 Evaluation 13
3.4.4 Limitations of Methodology 14

4 Research Plan **15**

5 Conclusion **16**

Chapter 1

Introduction

In modern healthcare, medical diagnoses are no longer determined by a doctor alone. Predictive models are increasingly being used to support medical decision making, but this raises a difficult question: Who is to blame when they get it wrong?

In medicine, a diagnosis must be more than an answer: it must be something that can be explained, questioned, and justified. If a model contributes to a missed diagnosis or harmful treatment recommendation, responsibility becomes difficult to assign if there is no reasoning to inspect and no logic to challenge. When we receive a diagnosis from a medical professional, we expect that they will be able to explain why a particular decision was made, what evidence supports it, and whether it agrees with accepted medical knowledge. If predictive models are used to support clinical decisions, they should be held to a similar standard.

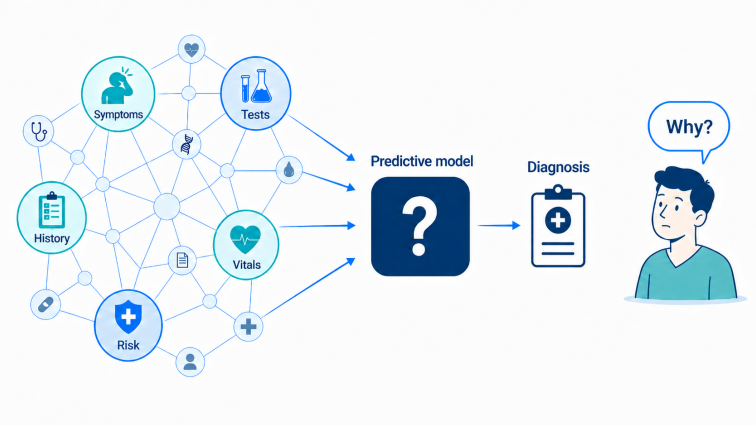


Figure 1.1: Many complex relationships between symptoms, measurements, tests, medical history etc. are captured by a 'black-box' predictive model which presents a diagnosis where the reasoning behind the diagnosis is unclear

Unfortunately, many powerful predictive models cannot clearly show their internal reasoning. It is difficult for a clinician to trust a conclusion when they have no way of knowing whether the model has learned a medically meaningful relationship, or whether it has relied on a pattern that is accidental, misleading, or inconsistent with clinical knowledge. In a low-risk setting, an unexplained prediction may be inconvenient, but in healthcare, it can affect treatment decisions, patient safety, and professional responsibility.

Yet predictive models remain invaluable in medical settings. Diagnosis is often shaped by many interacting factors, including symptoms, test results, medical history, medication, and lifestyle. It is unlikely that a clinician would be able to keep track of every possible relationship between these factors, especially in rare or complicated cases. A well-designed model can help identify patterns that may otherwise be missed. The aim, therefore, is not to replace clinical judgement, but to support it in a way that remains understandable and accountable.

Bayesian networks offer a promising way to approach this problem because they represent medical reasoning as a visible structure of relationships. They can show how different pieces of information are connected and how they contribute to a diagnosis. However, traditional Bayesian networks can struggle when medical relationships become highly complex. Real diagnosis often depends on many interacting factors, and these combined effects can be difficult to capture using simple relationship estimates.

This creates the central problem addressed in this research. Deep learning models are powerful enough to capture complex patterns, but they are often difficult to interpret. Bayesian networks are easier to understand, but their traditional learning methods may be too limited for complex medical relationships. To combine these strengths, this research investigates whether neural networks can be used within the interpretable structure of a Bayesian network to improve its ability to capture complex relationships.

The remainder of this proposal is organized as follows. Chapter 2 reviews existing work on interpretable medical prediction, Bayesian networks in healthcare, and approaches that combine predictive performance with explainability. Chapter 3 presents the research methodology, including the experimental design and evaluation approach. Chapter 4 outlines the research plan and timeline for completing the study.

Chapter 2

Related Work

A review is provided of research relevant to medical Bayesian networks that use neural networks to estimate parameters. Previous interpretable medical model implementations are considered, as well as common parameter estimation methods used for discrete Bayesian networks, including their benefits and drawbacks, and finally, some other approaches that combine interpretability and deep learning.

2.1 Conceptual Framework

In medical domains, predictive models must be transparent enough for clinicians to inspect, question, and justify their reasoning. Bayesian networks are particularly well-suited, as their graphical structure explicitly represents relationships between clinical variables. However, their practical limitation lies in their parameter estimation: traditional tabular-based methods are poorly suited to large, complex clinical settings. In contrast, deep learning models are capable of learning complex non-linear relationships between many variables, but are ill-suited to medical decision support due to their black-box nature.

This research proposes combining both models: using neural network's function approximation capability only for local parameter estimation, while retaining the interpretable graphical structure of the Bayesian network. Instead of representing each local conditional distribution as a separate table entry for every parent configuration, a CPD can be modelled as a learned function from parent variables to the probability distribution of a child variable. This allows parameters to be shared across related parent configurations, enabling more complex dependencies to be represented.

This approach is motivated by a Neural-Gaussian Network Model proposed in [Gaogle 2026], a dissertation submitted to the Advances in Data Analysis and Classification Journal, in which neural networks replace the conventional linear-Gaussian parameterization to estimate the CPDs of Gaussian Bayesian networks. This work demonstrates that neural parameter estimation within the interpretable structure of a Bayesian network is both feasible and competitive with traditional baselines.

However, the Gaussian setting assumes that all variables are continuous and jointly normally distributed, an assumption that is rarely satisfied in clinical practice. Although many physiological measurements are continuous, broader clinical reasoning uses discrete variables such as symptom presence or categorical risk levels, which are more common in medical datasets and more interpretable to clinicians who reason in terms of thresholds and categories rather than continuous probability densities. This research therefore aims to extends [Gaogle 2026] to the discrete setting, aiming to combine the interpretability of discrete Bayesian networks with the representational power of neural networks.

2.2 Interpretable Models for Medical Diagnosis

Model interpretability is a critical requirement in high-stakes medical environments. Clinicians must be able to understand and validate the reasoning behind model predictions to ensure safe and ethical decision making, and transparency in decision systems can be a requirement for legal and regulatory compliance [Hulsen 2023]. Consequently, transparent clinical decision support systems (CDSS) are becoming more widely implemented in health institutions to inform reliable medical decision-making. Depending on the models used, these systems can be classified as knowledge-based CDSS (KB-CDSS), and data-based CDSS (DB-CDSS)[Xu *et al.* 2023].

KB-CDSS focuses on representing a knowledge base elicited from medical expert knowledge. Initial implementations made use of decision rules [Christopher *et al.* 2015], defining 'if-then' rules which are easy to implement, but depend on exact conditions matching. More commonly used fuzzy logic models [El–Sappagh *et al.* 2018] handle uncertainty by allowing variables to take degrees of truth between 0 and 1. While these techniques create highly interpretable models, they can become difficult to manage and do not scale easily.

DB-CDSS makes use of large medical datasets to train prediction models. Interpretable DB-CDSS models include logistic regression and decision trees [Liu *et al.* 2020] [Lelis *et al.* 2020]. Logistic Regression estimates probabilities using weighted features, but fails to capture non-linear relationships and thus may under-perform on complex data. Decision trees are hierarchical rule-based models that make predictions through sequential splits, but can be prone to overfitting.

Bayesian networks (BNs) provide a compelling alternative by combining the strengths of both knowledge-based and data-driven approaches. Expert knowledge is incorporated through graph structure, while probabilistic relationships between variables are learned from data [Nzomo and Moodley 2025]. Because of their interpretable graphical structure and capacity to reason under uncertainty, BNs have been widely applied to medical diagnosis and risk prediction [Kyrimi *et al.* 2021][Zhong *et al.* 2024]. A scoping review on BNs in disease diagnosis and prognosis was conducted [Polotskaya *et al.* 2024] , finding a consistent pattern of high diagnostic accuracy across a range of clinical domains. However, BNs are still outperformed in predictive accuracy by black-box deep learning models such as gradient boosting ensembles.

2.3 Parameter Estimation for Discrete Bayesian Networks in Medical Contexts

Although widely researched, medical BN implementations rarely specify the parameter estimation methods used, Polotskaya *et al.* [2024], with many models defaulting to Maximum Likelihood Estimation (MLE) or Bayesian Parameter Estimation (BPE). However these standard approaches can be problematic over high dimensional, sparse, feature imbalanced medical datasets.

The fundamental challenge with MLE and BPE parameter estimation for discrete Bayesian networks, is the exponential growth of the conditional probability table with respect to parent in-degree[Koller and Friedman 2009]. A joint probability for every possible parent configuration must be stored in each CPT, meaning that even a moderate number of parent variables produces CPTs that are too large to estimate reliably from finite data. This scaling problem has a direct consequence for data sparsity: as the number of parent configurations grows exponentially, the available data is spread across an exponentially larger space of configurations, making many configurations unobserved or poorly represented even in large datasets[Oniśko *et al.* 2001].

In clinical settings, where data collection is expensive, and rare conditions are by definition underrepresented, this sparsity is especially acute[Oniśko and Drużdżel 2013]. Standard parameter estimation methods must therefore contend with both the representational burden of high-dimensional parent spaces and data insufficiency.

MLE estimates each CPT as a ratio of observed counts [Koller and Friedman 2009]:

$$\hat{\theta} = \frac{N[x_i, pa(x_i)]}{N[pa(x_i)]},$$

which performs well when data is abundant and complete, but fails in clinical datasets characterized by sparsity and class imbalance. When parent configurations are absent from the data, undefined estimates are produced, and rare clinical events are assigned a zero probability, preventing generalisation to unseen patient presentations.

These zero-valued parameters are a significant challenge when modelling medical data, as they propagate through Bayesian inference and permanently suppress otherwise valid diagnoses. A small amount of noise in numerical parameters will have a minimal impact on diagnostic accuracy as long as these zero-value parameters are avoided.[Oniśko and Drużdżel 2013].

BPE addresses these problems with the use of Dirichlet priors:

$$P(\theta|D) \propto P(D|\theta) \cdot P(\theta), \quad \theta \sim \text{Dir}(\alpha),$$

[Koller and Friedman 2009], which ensure non-zero probability for all state configurations. In medical domains these priors would ideally be informed by expert knowledge or clinical guidelines, but numerical estimates from clinicians are often unavailable or unreliable [Kyrimi *et al.* 2021].

MAP Estimation maximizes the posterior probabilities of the parameters:

$$\theta_{MAP} = argmax_{\theta} p(\theta|D),$$

[Koller and Friedman 2009], which avoids the complexity of full posterior distributions for high dimensional datasets, but simplifies to a single estimate which can result in biased prediction especially when data is sparse. To address class imbalance, CaMAP constructs an informative prior from expert-provided parameter constraints. CaMAP outperformed both MLE and MAP with uniform Dirichlet priors on a brain tumour BN with as few as 20 patient records[Di *et al.* 2021]

Clinical variables often have complex non-linear relationships that cannot always be captured with posterior distributions, especially if the given priors are not representative of the data. Noisy-OR gates have been used to model high-dimensional data with non-linear dependencies. They use an approximation to reduce CPT complexity from exponential to linear for binary variables, but at the cost of independence assumptions that do not hold clinically [Oniśko *et al.* 2001].

Therefore, it can be seen that parameter estimation for medical BNs remains underexplored and current methods fail to sufficiently address primary concerns with modeling medical data, including high dimensionality, sparsity, and class imbalance.

2.4 Hybrid Models for Interpretability and Predictive Accuracy

There is an increasing trend in medical diagnostics, of implementations that use machine learning models in conjunction with post-hoc explainability techniques, such as SHapley Additive exPlanations(SHAP) and Local Interpretable Model-agnostic Explanations(LIME)[Hossain *et al.* 2023][Kokkotis *et al.* 2022][El-Sofany *et al.* 2024]. SHAP quantifies the contribution of each feature towards the model's prediction, while LIME provides local explanations for individual predictions. These techniques have been used extensively to explain different black-box models in a variety of different medical fields, such as [Kokkotis *et al.* 2022], which uses a multilayer perceptron for stroke prediction, and [El-Sofany *et al.* 2024] which uses XGBoost for heart disease prediction.

However, post-hoc explanations of black-box models are increasingly criticized in high-stakes domains. These explanations are not necessarily faithful to the internal reasoning of the original model[Rudin 2019][Slack *et al.* 2019]. This can be problematic when they are used by clinicians to validate the reasoning behind a model's prediction.

The prevailing view in clinical AI presents model accuracy and interpretability as a fundamental trade-off. However, [Rudin 2019] argues that in many cases this is a result of poor model design rather than a fundamental constraint. One such demonstration is given in [Arif *et al.* 2025], which implements several baseline classifiers with SHAP and LIME post-hoc explanations for heart disease prediction, and the Decision Tree classifier outperforms traditional black-box methods such as Random Forest and Gradient Boosting.

Inherently interpretable models are better suited to clinical settings, as they provide accurate explanations for their predictions, and make it easier to detect possible bias. Generalized Additive Models (GAMs) are at the frontier of inherently interpretable models for tabular data. A GAM prediction is simply the sum of an independently learned function from each feature, allowing easy visualization of the influence of each feature:[Nori *et al.* 2019]

$$g(E[y]) = \beta_0 + \sum f_i(x_i).$$

Explainable Boosting Machines (EBMs) provide an extension of GAMs where each feature function f_i is learned with cyclic gradient boosting, which enables accuracy comparable to random forest and gradient boosting, while remaining fully interpretable [Nori *et al.* 2019].

EBMs have been shown to perform well on some medical datasets [Bosschieter *et al.* 2023], as they are able to model non-linear variable dependencies and handle missing data well. However, these successful implementations include datasets with a relatively low number of feature. EBMs do not perform well on high-dimensional datasets, as learning a separate boosted tree for every feature becomes intractable as the number of features increases [Greenwell *et al.* 2023].

2.5 Research Gap

Interpretable models such as BNs are well-suited to clinical reasoning and regulatory requirements, but their CPT-based parameter estimation methods (MLE, BPE, MAP) are insufficient to capture the non-linear, high-order dependencies present in real clinical data. More expressive estimation strategies either sacrifice the probabilistic transparency that makes BNs clinically useful, or depend on expert-provided priors that are impractical to specify at scale.

Deep learning models that can represent such complex dependencies are not interpretable by design, and post-hoc explanation methods can be unfaithful to the model, making them ill-suited in high-stakes settings. Inherently interpretable alternatives such as EBMs degrade on high-dimensional data.

Therefore a gap is identified in the current literature for a parameter learning approach that is simultaneously expressive enough to model complex clinical dependencies, robust to the sparsity and class imbalance characteristic of medical datasets, and compatible with the interpretable probabilistic structure of Bayesian networks.

Chapter 3

Methodology

3.1 Problem Statement

Bayesian Networks (BNs) provide an interpretable structure that is well-suited to modeling the probabilistic relationships between diseases, symptoms, and diagnostic test results in clinical settings. However, standard parameter estimation techniques present a fundamental representational limitation that prevents BNs from capturing the complex, non-linear relationships between clinical variables.

A Bayesian network is defined by a directed acyclic graph G, where each node has an associated conditional probability distribution (CPD), and the joint distribution factorizes according to G as:

$$P(X_1, ..., X_n) = \prod_{i=1}^{n} P(X_i | Pa_{X_i}),$$

where Pa_{X_i} denotes the parents of X_i in G [Koller *et al.* 2007].

In discrete BNs, methods such as Maximum Likelihood Estimation (MLE) and Bayesian Parameter Estimation (BPE) learn local conditional distributions $P(X_i|Pa_{X_i})$ independently using conditional probability tables (CPTs). The primary limitation arises from the conditioning on each parent Pa_{X_i}. As the number of parents increases, the number of parameters required to represent $P(X_i|Pa_{X_i})$ grows exponentially, such that a variable with k binary parents requires $2^{k+1} - 1$ parameters.

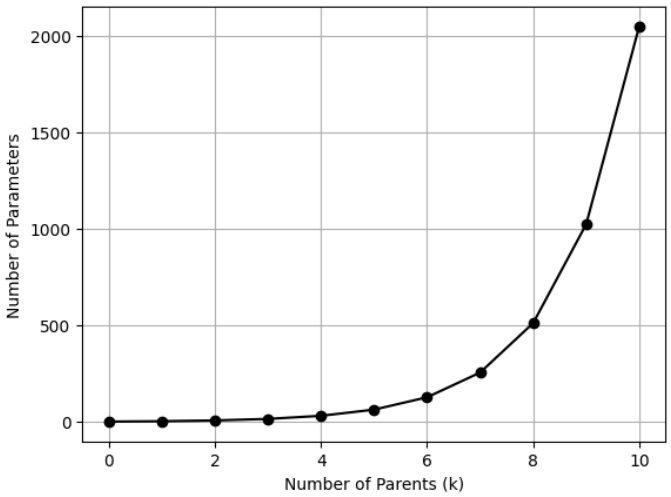


Figure 3.1: A graph that illustrates the exponential growth of CPT parameters

In medical domains, variables often exhibit complex relationships involving many interacting dependencies. The exponential parameter growth makes MLE and BPE impractical when variables have many dependencies, as the number of parameters to be estimated becomes infeasible. As a result, these standard parameter estimation methods are unable to effectively capture the complex, high-dimensional relationships that characterize real-world clinical data.

Although deep learning models are capable of capturing such complex dependencies, and have demonstrated strong predictive performance in medical diagnosis, their 'black-box' model design makes them impractical in clinical settings where medical professionals require transparent reasoning from diagnostic models. The challenge is therefore to develop a parameter learning approach for medical BNs that is able to capture non-linear variable relationships characteristic of real-world clinical data, without sacrificing the interpretability that makes BNs well-suited to clinical decision support.

3.2 Purpose Statement

This research aims to leverage the function approximation capabilities of neural networks, for parameter estimation in BNs for medical classification, thus enabling capture of complex relationships between clinical variables, while maintaining the interpretable structure and variable associations that make Bayesian Networks well-suited to clinical decision support.

3.3 Research Questions and Experimental Setup

RQ1: What are the representational limitations of MLE in capturing complex dependencies in a discrete Bayesian Network?

We construct a set of synthetic ground-truth Bayesian networks in which a target node is assigned a varying number of parent nodes. Samples are drawn from each ground-truth network and used for CPT parameter estimation with Maximum Likelihood Estimation.

KL Divergence is used to compare each Bayesian Network's estimated distribution with its corresponding ground truth-network. For each in-degree value, we gradually increase the network size of the ground-truth networks, until the model can no longer approximate the ground truth distribution reliably because the computation becomes infeasible.

The experiment therefore measures how the KL divergence changes as both network size and parent in-degree increase, continuing until a point of failure is reached. This allows the experiment to identify the point at which standard CPT parameter estimation becomes unreliable as parent complexity increases.

RQ2: Can we replace the local probability model with a deep neural network in each family distribution in the Bayesian network in order to better capture more complex dependencies?

Using the same ground-truth distributions as in RQ1, the tabular conditional probability distribution (CPT) associated with each node is replaced with a neural-network-based conditional probability distribution (CPD). For each node X_i, a neural network is trained to approximate the local conditional distribution $P(X_i \mid Pa(X_i))$. The neural network uses a softmax activation function at the output layer, which enables it to produce a valid probability distribution over the discrete states of X_i.

KL divergence is used to compare the learned neural CPDs against the corresponding ground-truth conditional distributions. In order to determine whether neural-network-based parameter estimation remains reliable at higher parent in-degrees where traditional CPT-based estimation begins to fail, we measure how the KL divergence changes as both network size and parent in-degree increase, in the same manner as RQ1.

The neural-network-based Bayesian Network is then compared directly against the CPT-based Bayesian Network trained using MLE. This allows us to evaluate whether neural-network-based CPDs are able to estimate high-dimensional conditional distributions more effectively than standard CPT-based parameter estimation.

RQ3: Can neural CPDs model real clinical variables with more parent configurations than CPT-based estimation permits?

RQ3 tests whether the failure identified in RQ1 and RQ2 is observable in practice, using the BRFSS Diabetes Health Indicators dataset and the causal Bayesian network published by [Zahoor *et al.* 2024]. Where the synthetic experiments vary the number of parent configurations through in-degree, the graph here is taken as given, so the same quantity is varied through variable cardinality.

The same graph is fitted at both the full cardinalities available in the data, and at reduced cardinalities which are used in practice to reduce CPT growth. The held-out predictive log-likelihood is compared across estimators against the ratio of parent configurations to available samples. If neural CPDs maintain accuracy where CPT-based estimators degrade, this demonstrates on real data that the reduction is not necessary.

3.4 Detailed Methodology

3.4.1 Data Generation and Pre-processing

As the objective of this study is to evaluate parameter estimation performance under controlled conditions, synthetic datasets will be generated from known ground-truth Bayesian Networks. This allows the true conditional probability distributions to be known in advance, enabling direct comparison between the learned distributions and the true generating distributions.

This direct comparison would not be possible with real-world medical data, where the true underlying conditional distributions of such data are unknown. The synthetic network configurations are designed to reflect the structural properties (large network size, sparsity and high parent in-degree) that characterize real clinical datasets, ensuring that findings are relevant to the intended application domain.

Ground-Truth Generation

A collection of ground-truth discrete Bayesian Networks will be generated from structures of varying complexity. A target node X_i will be assigned a predefined number of parent variables, $Pa(X_i) = \{X_1, X_2, ..., X_k\}$, where k denotes the parent in-degree.

Table 3.1 specifies the ground truth scenarios that will be used, with varying network size, graph density, and parent in-degree. This enables the evaluation of parameter estimation methods over increasingly complex density and parent in-degree structure configurations, and how this scales as the network size increases.

The sparse low-in-degree scenario provides a baseline in which CPTs remain small and MLE is expected to recover the ground-truth distribution reliably, while the moderate and dense scenarios are used to observe how estimation performance changes as parent configurations increase. The sparse high-in-degree target scenario isolates the effect of local CPD complexity while keeping the overall graph sparse.

Table 3.1: Ground-truth Bayesian network scenarios used for experimental evaluation

Ground-truth scenario	**Number of nodes**	**Density range**	**In-degree range**
Sparse / low in-degree	5, 10, 20, 30, 50	Sparse / tree-like (0.05–0.20)	1–3
Moderate density / medium in-degree	10, 20, 30, 50	Moderately dense, (0.20–0.50)	3–6
Dense / high in-degree	20, 30, 50	Dense / near fully connected (0.50–1.00)	6–10 across multiple nodes
Sparse / high-in-degree target	20, 30, 50	Sparse overall with dense target family (0.05–0.25)	6–10 for selected target nodes

Synthetic Data Sampling

Synthetic datasets will be generated through ancestral sampling from each ground-truth Bayesian Network. To reduce variance caused by random sampling, multiple independent datasets will be generated per experimental condition using different random seeds, where each dataset is sampled independently from the same ground-truth network.

To reflect the data scarcity characteristic of real clinical settings, the number of training samples per experimental condition will be held constant as parent in-degree increases. This ensures that any observed degradation in parameter estimation performance is attributable to the growing complexity of the conditional probability space rather than variation in data availability, mirroring the challenge faced in practice where clinical data does not scale with model complexity.

Since the true ground truth distributions are known exactly, learned parameters can be evaluated directly against the true CPTs without requiring a held-out test set. For CPT-based MLE, each of the independently generated datasets is used in full for parameter estimation. For neural-network-based CPDs, a small validation subset is held out from each dataset prior to training. This subset is used exclusively for early stopping and hyperparameter selection.

The same synthetic datasets and ground truth network structures will be used across all parameter estimation methods to ensure a fair comparison.

Real-World Application Data

The real-data component uses the BRFSS Diabetes Health Indicators dataset, comprising 253,680 survey responses across 21 health-related variables and a binary diabetes indicator. Because responses are collected by structured survey instrument, every variable except body mass index is natively categorical, so the discretisations compared in RQ3 are defined by the survey instrument rather than chosen here. The finer setting uses the categories as collected, while the coarser reproduces the collapsing applied in practice to keep conditional probability tables estimable.

Estimator performance is assessed by k-fold cross-validation, with held-out negative log-likelihood averaged across folds and the standard deviation reported. Within each training fold a further validation split is held out for early stopping and hyperparameter selection, so that no test fold influences model selection. Training subsets of varying size are drawn within folds, allowing the ratio of parent configurations to available samples to be varied independently of discretisation granularity. Records with missing values are excluded and the proportion removed is reported.

BRFSS is publicly available, fully de-identified, and exempt from institutional review board oversight. MIMIC, should it be used as a secondary application, is similarly de-identified but requires PhysioNet credentialing and a data use agreement, a process of several weeks on which the present design does not depend.

3.4.2 Methods

Two parameter estimation methods are implemented and evaluated: Maximum Likelihood Estimation as the traditional baseline, and neural-network-based conditional probability distributions as the proposed alternative.

Maximum Likelihood Estimation

For the traditional Bayesian Network experiments, CPT parameters are estimated using Maximum Likelihood Estimation (MLE). For a variable X_i with parent configuration u_j, the MLE estimate is defined as:

$$\hat{P}(X_i = x_k \mid Pa(X_i) = u_j) = \frac{N_{ijk}}{N_{ij}},$$

where N_{ijk} is the number of samples in which $X_i = x_k$ and $Pa(X_i) = u_j$, and N_{ij} is the number of samples in which $Pa(X_i) = u_j$ [Koller and Friedman 2009].

Neural Network Parameter Estimation

To address the scalability limitations of tabular CPT parameterization, this research replaces the local conditional probability table with a neural-network-based conditional probability distribution (CPD).

For each node X_i, a neural network f_θ is trained to approximate $P(X_i \mid Pa(X_i))$, where the parent variables $Pa(X_i)$ are provided as input features to the network. The neural CPD therefore learns a functional mapping:

$$P(X_i \mid \mathrm{Pa}(X_i)) = f_\theta(\mathrm{Pa}(X_i)),$$

parameterized by neural network weights θ.

The neural network consists of fully connected feedforward layers. Given an input vector x representing the parent configuration, the hidden representation is computed as $h = \phi(W_1 x + b_1)$, where W_1 represents the layer weight matrix, b_1 represents the bias vector, and ϕ denotes the activation function [Goodfellow *et al.* 2016].

The output layer uses a softmax activation function to produce a valid probability distribution over the discrete states of X_i: [Goodfellow *et al.* 2016]

$$\hat{y} = \mathrm{softmax}(W_2 h + b_2).$$

This allows the neural network to approximate discrete conditional probability distributions while sharing parameters across parent configurations.

Neural Network Model

The neural CPDs are trained using cross-entropy loss, defined as:

$$\mathcal{L}(\theta) = -\sum_{i=1}^{N} \log P_\theta(x_i \mid pa_i),$$

where N is the number of training samples, x_i is the observed node state, and pa_i is the observed parent configuration [Goodfellow *et al.* 2016].

Multiple neural network structures and hyperparameter values are evaluated as neural CPDs, in order to evaluate which neural network configurations are most effective for approximating conditional probability distributions as dimensionality increases.

Table 3.2: Neural network parameter configurations for neural CPD estimation

Neural Network	**Architecture**	**Hidden activation**	**Regularization**	**Training parameters**
Linear Softmax	No hidden layer; direct mapping from parent variables to child-state probabilities	None	L2 weight decay: 10^{-4}	LR 10^{-3}; batch size 64; 100 epochs
Shallow MLP	1 hidden layer with 32 hidden units	ReLU	Dropout: 0.10; L2 weight decay: 10^{-4}	LR 10^{-3}; batch size 64; 150 epochs
Medium MLP	2 hidden layers with 64 and 32 hidden units	ReLU	Dropout: 0.20; L2 weight decay: 10^{-4}	LR 10^{-3}; batch size 64; 200 epochs
Deep MLP	3 hidden layers with 128, 64, and 32 hidden units	ReLU or tanh	Dropout: 0.30; L2 weight decay: 10^{-3}	LR 5×10^{-4}; batch size 64; 250 epochs

Table 3.2 specifies four neural network configurations ranging from a minimal linear softmax baseline with no hidden layers, to a deep MLP (Multi-Layer Perceptron) with three hidden layers. This enables the evaluation of whether increased representational capacity improves conditional probability estimation as parent in-degree grows. All configurations use a softmax output layer to ensure a valid probability distribution over the discrete states of the child node. Input representation is also evaluated by comparing one-hot encoded parent configurations against learned embedding representations.

3.4.3 Evaluation

Synthetic Experiments

The primary evaluation metric used in this study is Kullback-Leibler (KL) divergence, which measures the difference between the learned conditional probability distributions and the known ground-truth distributions used to generate the synthetic datasets. For each experiment, the learned Bayesian Network parameters are compared against the corresponding ground-truth conditional distributions using:

$$D_{KL}(P^*||\hat{P}) = \sum_x P^*(x) \log \frac{P^*(x)}{\hat{P}(x)},$$

where P^* denotes the ground-truth probability distribution and $\hat{P}$ denotes the learned distribution [Koller and Friedman 2009].

KL divergence will be plotted against increasing network size for multiple parent in-degree values. Separate curves will be generated for each in-degree setting, allowing direct comparison of parameter estimation reliability of the standard method and proposed neural method.

Real-Data Experiments

Since the ground-truth conditional distributions are unavailable, predictive performance is assessed by negative log-likelihood per observation on held-out data. The expected held-out negative log-likelihood is the cross-entropy between the true and estimated distributions,

$$H(P^*, \hat{P}) = H(P^*) + D_{KL}(P^* \,\|\, \hat{P}),$$

[Koller and Friedman 2009] and since $H(P^*)$ does not depend on the estimator, differences in negative log-likelihood are exactly differences in KL divergence. The ranking of estimators and the gaps between them therefore transfer from the synthetic setting, while the absolute divergence from truth does not, since the entropy offset is unknown.

Computational Metrics

For synthetic and real-data experiments computational scalability metrics including training time, memory usage, and parameter count will be recorded to evaluate how the computational requirements of each parameter estimation method scale as complexity and dimensionality increases.

Each experimental configuration is repeated across multiple independently generated datasets or resamples, and the mean and standard deviation are reported to reduce the effect of random sampling variation.

3.4.4 Limitations of Methodology

The synthetic experiments permit precise measurement of estimation accuracy but cannot on their own establish relevance to clinical practice; the real-data component addresses this, but introduces limitations of its own. On real data the true conditional distributions are unknown, so evaluation necessarily shifts to predictive fit, which identifies relative differences between estimators but not absolute distance from truth. The network structure is adopted from prior work rather than learned or independently validated

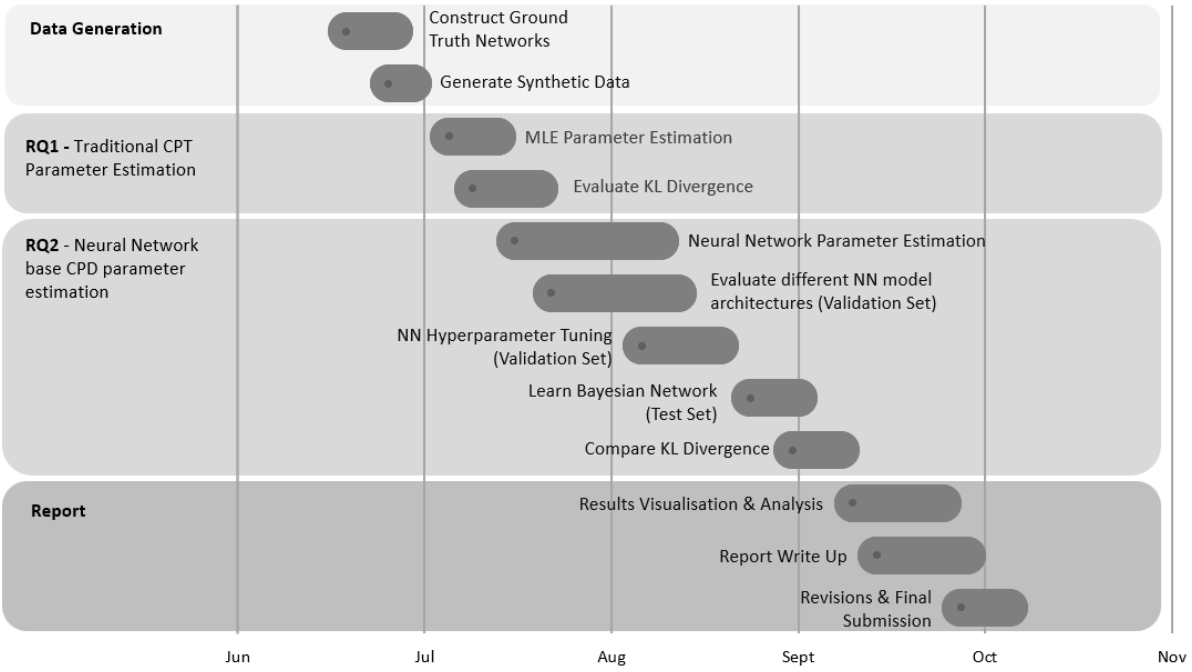
Another limitation is that the experiments are restricted to discrete Bayesian Networks. Continuous variables must therefore be discretized before being represented within the Bayesian Network framework, which may result in information loss and reduced representational fidelity.

Finally, while neural-network-based CPDs may improve scalability in high-dimensional parent spaces, they reduce the direct interpretability of traditional conditional probability tables and introduce greater computational complexity and hyperparameter tuning requirements.

Chapter 4

Research Plan

The research is organised into three sequential phases, as illustrated in the Gantt chart below. The first phase covers the construction of ground-truth Bayesian networks and synthetic data generation. The second phase addresses the two research questions, beginning with MLE-based CPT parameter estimation for RQ1, followed by the implementation and evaluation of neural-network-based CPDs for RQ2. The third phase covers results analysis, report writing, and final submission. The three phases are planned to run from June to November 2026. This plan can be extended to include a real-data application phase addressing RQ3, comprising data preprocessing, estimator comparison across discretisation levels, and analysis.



A potential risk is that hyperparameter tuning across four neural network architectures and multiple ground-truth network scenarios may require more computational resources and time than anticipated, compressing the time available for report writing. To mitigate this, hyperparameter tuning will be prioritized on smaller network configurations first, and early stopping will be used to limit unnecessary training time.

Chapter 5

Conclusion

Medical diagnosis is a high-stakes domain where predictive models must be not only accurate but transparent. A clinician who cannot inspect or question a model's reasoning cannot safely act on it, and cannot bear professional responsibility for decisions it influences. Bayesian networks are among the most promising frameworks for this setting, because their graphical structure makes probabilistic reasoning visible and inspectable. Yet this promise is undermined by a fundamental limitation: standard parameter estimation methods cannot reliably capture the complex, high-dimensional dependencies that characterise clinical data, particularly under the sparsity and class imbalance common in medical datasets.

This proposal addresses that limitation directly. By replacing traditional conditional probability tables with neural-network-based conditional probability distributions, the proposed approach retains the interpretable graphical structure of Bayesian networks while leveraging the representational power of deep learning for local parameter estimation. The research is grounded in experimental design using controlled synthetic ground-truth networks, enabling precise evaluation of parameter estimation performance across increasing levels of network complexity, through the use of KL divergence as a principled comparison metric.

The contribution of this work is therefore twofold. First, it provides an analysis of where and how standard parameter estimation fails in discrete Bayesian networks. Second, it proposes and evaluates a neural parameter estimation framework, that is simultaneously expressive, robust to the sparsity and high-dimensionality common in medical settings, and compatible with the interpretable structure that makes Bayesian networks suitable for clinical decision support. In doing so, this research addresses a longstanding tradeoff in clinical AI between predictive power and model transparency, moving towards the development of diagnostic models that are both accurate and explainable.

References

[Arif *et al.* 2025] Muhammad Arif, Fatema Tuz Johora, Abu Salyh Muhammad Mussa, and Md. Sajaratul Yakin Sagor. Explainable artificial intelligence for heart disease prediction: A shap and lime-based approach. *2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*, 2025.

[Bosschieter *et al.* 2023] Tomas M. Bosschieter, Zifei Xu, Hui Lan, Benjamin J. Lengerich, Harsha Nori, Ian Painter, Vivienne Souter, and Rich Caruana. Interpretable predictive models to understand risk factors for maternal and fetal outcomes. *Journal of Healthcare Informatics Research*, 2023.

[Christopher *et al.* 2015] Jabez Christopher, H. Khanna Nehemiah, and A. Kannan. A clinical decision support system for diagnosis of allergic rhinitis based on intradermal skin tests. *Computers in Biology and Medicine*, 2015.

[Di *et al.* 2021] Ruo-hai Di, Peng Wang, Chuchao He, and Zhi-gao Guo. Constrained adjusted maximum a posteriori estimation of bayesian network parameters. *Entropy*, 2021.

[El-Sofany *et al.* 2024] Hosam F. El-Sofany, Belgacem Bouallègue, and Yasser M. Abd El-Latif. A proposed technique for predicting heart disease using machine learning algorithms and an explainable ai method. *Scientific Reports*, 2024.

[El–Sappagh *et al.* 2018] Shaker El–Sappagh, José M. Alonso, Farman Ali, Amjad Ali, Jun-Hyeog Jang, and Kyung Sup Kwak. An ontology-based interpretable fuzzy decision support system for diabetes diagnosis. *IEEE Access*, 2018.

[Gaogle 2026] Aashna Gaogle. *Using Neural Networks to Estimate the Parameters of Gaussian Bayesian Networks*. Master's thesis, University of the Witwatersrand, Johannesburg, South Africa, 2026. Unpublished Masters dissertation.

[Goodfellow *et al.* 2016] Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning*. MIT Press, 2016.

[Greenwell *et al.* 2023] Brandon M. Greenwell, Annika Dahlmann, and Saurabh Dhoble. Explainable boosting machines with sparsity - maintaining explainability in high-dimensional settings. *arXiv.org*, 2023.

[Holzinger *et al.* 2019] Andreas Holzinger, Georg Langs, Helmut Denk, Kurt Zatloukal, and Heimo Müller. Causability and explainability of artificial intelligence in medicine. *Wiley Interdisciplinary Reviews-Data Mining and Knowledge Discovery*, 2019.

[Hossain *et al.* 2023] Md. Imran Hossain, Ghada Zamzmi, Peter R. Mouton, Md Sirajus Salekin, Yu Sun, and Dmitry B. Goldgof. Explainable ai for medical data: Current methods, limitations, and future directions. *ACM Computing Surveys*, 2023.

[Hulsen 2023] Tim Hulsen. Explainable artificial intelligence (xai): Concepts and challenges in healthcare. *Applied Informatics*, 2023.

[Kokkotis *et al.* 2022] Christos Kokkotis, Georgios Giarmatzis, , Serafeim Moustakidis, Themistoklis Tsatalas, Dimitrios Tsiptsios, onstantinos Vadikolias, and Nikolaos Aggelousis. An explainable machine learning pipeline for stroke prediction on imbalanced data. *Diagnostics*, 2022.

[Koller and Friedman 2009] Daphne Koller and Nir Friedman. *Probabilistic graphical models: principles and techniques*. MIT press, 2009.

[Koller *et al.* 2007] Daphne Koller, Nir Friedman, Lise Getoor, and Ben Taskar. Graphical models in a nutshell. *Introduction to statistical relational learning*, 43:1359–1366, 2007.

[Kyrimi *et al.* 2021] Evangelia Kyrimi, Scott McLachlan, Kudakwashe Dube, Mariana R. Neves,

Ali Fahmi, and Norman Fenton. A comprehensive scoping review of bayesian networks in healthcare: Past, present and future. *Artificial Intelligence in Medicine*, 2021.

[Lelis *et al.* 2020] Viviane M. Lelis, Eduardo Guzmán, and María-Victoria Belmonte. Non-invasive meningitis diagnosis using decision trees. *IEEE Access*, 2020.

[Liu *et al.* 2020] Ying Liu, Zhixiao Wang, Jingjing Ren, Yu Tian, Min Zhou, Tianshu Zhou, Kangli Ye, Yinghao Zhao, Yunqing Qiu, and Jingsong Li. A covid-19 risk assessment decision support system for general practitioners: Design and development study. *Journal of Medical Internet Research*, 2020.

[Nori *et al.* 2019] Harsha Nori, Samuel Jenkins, Paul Koch, and Rich Caruana. Interpretml: A unified framework for machine learning interpretability. *arXiv: Learning*, 2019.

[Nzomo and Moodley 2025] Mbithe Nzomo and Deshendran Moodley. Integrating knowledge graphs and bayesian networks: A hybrid approach for explainable disease risk prediction. *arXiv.org*, 2025.

[Oniśko and Drużdżel 2013] Agnieszka Oniśko and Marek J. Drużdżel. Impact of precision of bayesian network parameters on accuracy of medical diagnostic systems. *Artificial Intelligence in Medicine*, 2013.

[Oniśko *et al.* 2001] Agnieszka Oniśko, Marek J. Drużdżel, and Hanna Wasyluk. Learning bayesian network parameters from small data sets: application of noisy-or gates. *International Journal of Approximate Reasoning*, 2001.

[Polotskaya *et al.* 2024] Kristina Polotskaya, Carlos Segundo Muñoz-Valencia, A. Rabasa, José A. Quesada, Domingo Orozco-Beltrán, and Xavier Barber. Bayesian networks for the diagnosis and prognosis of diseases: A scoping review. *Machine Learning and Knowledge Extraction*, 2024.

[Rudin 2019] Cynthia Rudin. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 2019.

[Slack *et al.* 2019] Dylan Slack, Sophie Hilgard, Emily Jia, Sameer Singh, and Himabindu Lakkaraju. Fooling lime and shap: Adversarial attacks on post hoc explanation methods. *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, 2019.

[Xu *et al.* 2023] Qian Xu, Wenzhao Xie, Bolin Liao, Chao Hu, Lu Qin, Zhengzijin Yang, Huan Xiong, Lyu Yi, Yue Zhou, and Aijing Luo. Interpretability of clinical decision support systems based on artificial intelligence from technological and medical perspective: A systematic review. *Journal of Healthcare Engineering*, 2023.

[Zahoor *et al.* 2024] Sheresh Zahoor, Anthony C. Constantinou, Tim M Curtis, and Mohammed Hasanuzzaman. *Investigating the validity of structure learning algorithms in identifying risk factors for intervention in patients with diabetes*, 2024.

[Zhong *et al.* 2024] Lu Zhong, Fan Yang, Shanshan Sun, Lijie Wang, Hong Yu, Xiushan Nie, Ailing Liu, Ning Xu, Lanfang Zhang, Mingjuan Zhang, Yue Qi, Huaijun Ji, Guiyuan Liu, Huan Zhao, Yinan Jiang, Jingyi Li, Chengcun Song, Xin Yu, Liu Yang, Jinchao Yu, Hu Feng, Xiaolei Guo, Fujun Yang, and Fuzhong Xue. Predicting lung cancer survival prognosis based on the conditional survival Bayesian network. *BMC Medical Research Methodology*, 24:16, 2024.